

Differential Solvers in Computational Electromagnetics

"Computational electromagnetics, computational electrodynamics or electromagnetic modeling is the process of modeling the interaction of electromagnetic fields with physical objects and the environment. It typically involves using computationally efficient approximations to Maxwell's equations and is used to calculate antenna performance, electromagnetic compatibility, radar cross section and electromagnetic wave propagation when not in free space," according to Wikipedia

VISHNU GAUTAM

Remember yourself calculating electric field intensity, current density or vector potential in idealised cases during your 12th standard. Those problems really gave a summary of the electromagnetics (EM) working in different situations. Advancing further you may recall having calculated the field quantities in case of dipole antennae (standard case). Well these problems were easy to analyse because of the symmetry existing in those situations: Just apply the Maxwell equations and symmetry will do its work.

But what if the problem's geometry is not symmetric, what if there is an urgent need to find the parameters without going much into the mathematics, what if a practical situation (for example, studying effects of mobile phone on pacemaker) needs to be analysed in a couple of hours?

Surely calculating each Maxwell equation and approximating them will not be a fast solution to the situation from industry point of view. In these situations we use electromagnetic modelling or, precisely, computational electromagnetics.

Computational electromagnetics (CEM) is a very challenging and multidisciplinary science that aims at finding solutions of Maxwell equations at boundary conditions (broadly the material boundaries of geometry). It involves numerical methods, efficient (not effective) algorithms and their

computer implementations (see Fig. 1).

The CEM implementation should have following characteristics:

Accuracy. The results should be accurate to the real situations.

Stability. Results should be stable, which means, if the parameters and excitation are changed then its output

should be predictable, that is, it should not become unstable.

Scalability. It should be scalable in terms of the computer memory. If it consumes a large memory then surely it

would be unacceptable.

Flexibility. Generally, one method produces best result for one type of problem. The implementation should be made flexible so that it would produce stable results for other type of problems as well.

Parallelisation. For fast results, one computer is not enough. So it must offer parallel computing.

The computing cost depends on size of the solution and time required to find the solution. It should be as low as possible. At the core of computational electromagnetics is the numerical method used for a specific problem. Broadly, there are two types of methods for solving EM problems:

Analytical method. This gives an insight into the process but is generally limited to idealised processes only.

Numerical method. This gives an approximate solution of any type of problem and structure. With numerical method the analysis and prototyping of complex systems is easy. It also allows investigation of material properties without performing any experiment, besides easy optimisation and modification for its implementation. Presently, simulators use a combination of both the methods to reduce the computational cost.

Implementation of CEM (Fig. 2) efficiently is not an easy task. There are various problems associated with their implementation, such as the type of material to be analysed (homogeneous/inhomogeneous, isotropic/anisotropic, etc), type of geometry

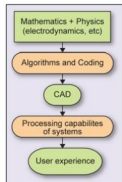


Fig. 1: Block diagram of CEM implementation (basic abstraction)

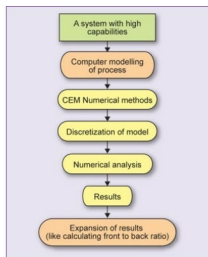


Fig. 2: CEM process